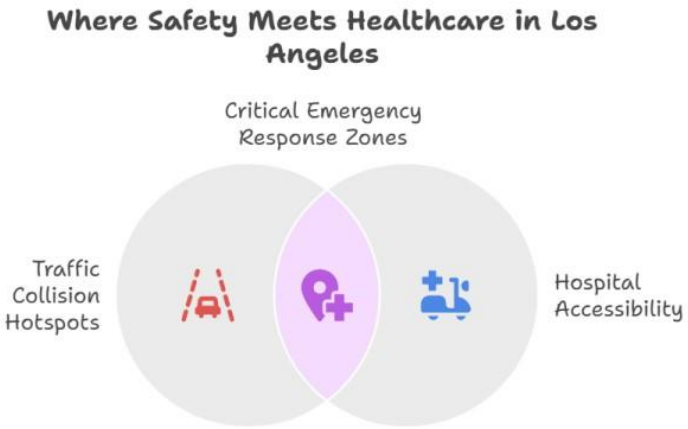
SAVING SECONDS, SAVING LIVES: "GEOSPATIAL ANALYSIS OF TRAFFIC COLLISION HOTSPOTS AND HOSPITAL ACCESSIBILITY IN LOS ANGELES"

DA 591 DATA SCIENCE AND BUSINESS ANALYTICS PRACTICUM

FINAL REPORT



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ABSTRACT

Traffic collisions are a major public safety concern in Los Angeles, where both injury frequency and emergency response demands remain high. This project integrates a large-scale collision dataset with statewide hospital locations to identify hotspots, assess hospital accessibility, and engineer features for a multi-stage crash severity model. Using geospatial methods, distance modeling, and behavioral indicators, the analysis highlights uneven access to emergency care and identifies key factors associated with severe outcomes. These findings provide a foundation for the predictive modeling phase and support data-driven safety planning.

Keywords — Traffic safety, geospatial analytics, hospital proximity, severity modeling.

INTRODUCTION

Los Angeles consistently experiences a high volume of traffic collisions, placing significant strain on emergency services and transportation systems. Understanding where collisions cluster, how quickly victims can reach hospitals, and which conditions contribute to injury severity is essential for improving safety outcomes.

This project integrates collision and hospital datasets to create a unified spatial and behavioral analysis. The results support long-term public safety goals and provide the analytical foundation for a crash-severity prediction model.

BACKGROUND INFORMATION

Urban mobility in Los Angeles spans complex road networks, diverse neighborhoods, and varying traffic conditions. Traffic behaviors such as speeding, DUI, hit-and-run, and nighttime driving are major contributors to severe injuries. Hospital accessibility varies across divisions, with some areas benefiting from dense trauma center coverage while others remain underserved. Geospatial analysis and predictive modeling offer a framework to explore these systemic challenges and propose data-driven improvements.

PROBLEM STATEMENT

Traditional collision analyses often focus only on counts or basic severity, without examining the combined effect of behavior, environment, and hospital access. This project addresses the following questions:

1. Where are the most prominent collision hotspots in Los Angeles?
2. Do these hotspots have timely access to emergency hospitals?
3. Can severity be predicted using spatial, temporal, and behavioral indicators?

The goal is to identify underserved regions, reveal high-risk patterns, and prepare a modeling-ready dataset for future predictive analysis.

DATASET DESCRIPTION

Traffic Collision Dataset:

The LAPD dataset comprises roughly 621,000 records of collisions, featuring vital information such as the date and time, coordinates, severity level, collision type, and basic victim demographics. Given that the data is geocoded, it facilitates the detection of hotspots and analysis of temporal trends on a daily, weekly, and seasonal basis.

Hospital Dataset:

The hospital dataset contains around 7,600 facilities in California, showcasing attributes like trauma level, availability of helipads, bed capacity, and geographic coordinates. After filtering to focus solely on hospitals within California, the dataset becomes ideal for assessing emergency access in relation to collision sites.

DATA CLEANING

Traffic Data Cleaning:

Location fields were transformed into accurate latitude and longitude values, enabling dependable distance computations and hotspot clustering. The date and time were combined into one DateTime field and divided into year, month, hour, and weekday, minimizing redundancy and enhancing consistency. Demographic fields regarding victims were normalized, and missing ages were filled in using the median value. MO Codes were extracted from the text, cleaned, and rearranged into key behavioral categories to develop useful analytical features.

Several columns were eliminated due to their contribution of noise, duplication of existing information, or lack of significance to the analysis. Free-text addresses were discarded since precise coordinates were already available. Administrative fields like "Date Reported" were removed as they did not reflect the actual time of the crash. Sparse or inconsistently coded fields were also taken out to simplify the dataset. The primary aim was to keep only those fields that enhanced spatial, temporal, or behavioral accuracy.

Hospital Data Cleaning:

Cleaning the hospital dataset began with converting projected coordinate values into standardized latitude and longitude to ensure compatibility with the collision dataset. Records with missing or invalid coordinates were removed to maintain reliability in spatial analysis. Trauma-level descriptions and helipad fields were normalized to maintain consistent categorization across all hospitals. Only facilities located in California were retained, as this study focuses entirely on Los Angeles accessibility patterns.

A number of fields were removed because they were irrelevant to emergency-accessibility evaluation. Administrative identifiers, metadata fields, contact information, and staffing counts did not influence distance calculations or trauma readiness and would only add unnecessary noise to the dataset. The final dataset preserves the essential variables needed to identify hospitals, understand their emergency capacity, and calculate proximity

Final Dataset Shapes

After data cleaning, transformation, and feature reduction, the project produced two streamlined, analysis-ready datasets:

The **traffic dataset** retains full event coverage while removing redundant and noisy fields, leaving 18 high-value features that support temporal analysis, hotspot detection, and severity prediction.

The **hospital dataset** is reduced from 36 to 8 essential attributes, sufficient to evaluate proximity, trauma capacity, and EMS accessibility without extraneous administrative fields.

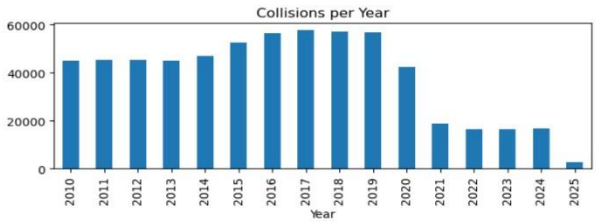
Accident–Hospital Distance Calculation

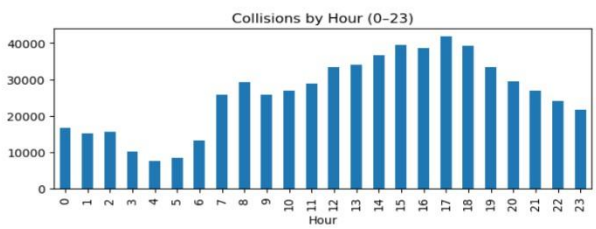
To assess emergency accessibility in Los Angeles, traffic accident data were paired with the closest hospitals using a geospatial method. First, the coordinates of accidents and hospitals were converted into radians to account for the curvature of the Earth. The Haversine formula was then utilized, with the Earth's radius set at 6,371 km, to convert angular differences into accurate straight-line distances measured in kilometers. To manage the extensive number of geospatial points efficiently, the BallTree nearest-neighbor algorithm was used. This approach created a combined dataset that paired each accident with its closest hospital, forming the basis for analyzing accessibility and coverage, while also producing two new columns: nearest_hosp_km, which indicates the distance in kilometers, and nearest_hosp_name, which names the corresponding facility. Results revealed

that although most accidents took place within 2–3 km of a hospital, ensuring quick access to emergency services, a considerable portion occurred more than 10 km away, pointing to underserved areas where longer response times could heighten the risk of fatal incidents.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) serves as an essential phase in comprehending the dataset's features and structure before implementing more advanced modeling methods. In this study, EDA was performed to uncover significant patterns, trends, and anomalies in the traffic collision data of Los Angeles and assess hospital accessibility. By exploring temporal trends (year, month, hour), spatial distributions (collision hotspots and distances to hospitals), and contextual factors (collision type, pedestrian involvement), EDA yields critical insights into the timing, locations, and reasons for accidents. These insights aid in feature engineering and model development, as well as highlight areas with inadequate hospital coverage, facilitating more effective emergency preparedness and policy planning.





Yearly Trends

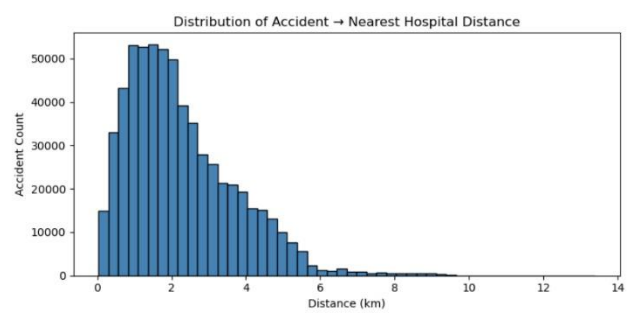
Analysis of yearly collision counts revealed a sustained high frequency of accidents between 2010 and 2014, followed by a noticeable peak from 2015 to 2018, when nearly 60,000 crashes occurred annually. After 2019, the data shows a sharp decline in collisions, which can be attributed in part to the COVID-19 pandemic and the resulting reduction in travel, commuting, and public mobility. This trend highlights how external factors, such as public health emergencies, can directly influence traffic safety dynamics.

Hourly Patterns

Collision counts followed a clear daily cycle. The lowest rates occurred in the early morning hours (3–6 AM), a period when overall traffic volumes are minimal. Fewer vehicles are on the road, and many traffic signals switch to night mode operations (flashing or extended cycles), reducing congestion-related collisions. However, while accident frequency is low during these hours, late-night and early-morning crashes are often more severe, influenced by factors such as driver fatigue, impaired driving, and higher average vehicle speeds on emptier roads.In contrast, the highest accident volumes were observed between 3–6 PM, coinciding with afternoon and evening rush-hour congestion. The combination of heavy traffic density, frequent stop-and-go conditions, and increased commuter activity contributes to elevated collision risk during these periods.

Hospital Accessibility

The geospatial analysis of collision distances to hospitals indicated that the majority of accidents occurred within 2–3 km of an emergency facility, suggesting generally good accessibility across much of Los Angeles. However, a significant subset of crashes occurred more than 10 km away from the nearest hospital, exposing coverage gaps where delayed emergency response times may increase the risk of fatalities or poor recovery outcomes. These findings underscore the need for strategic EMS staging and hospital resource allocation in underserved divisions.




```
count    620712.000000
mean          2.248997
std           1.451583
min           0.028000
25%           1.152000
50%           1.945000
75%           3.077000
max          13.394000
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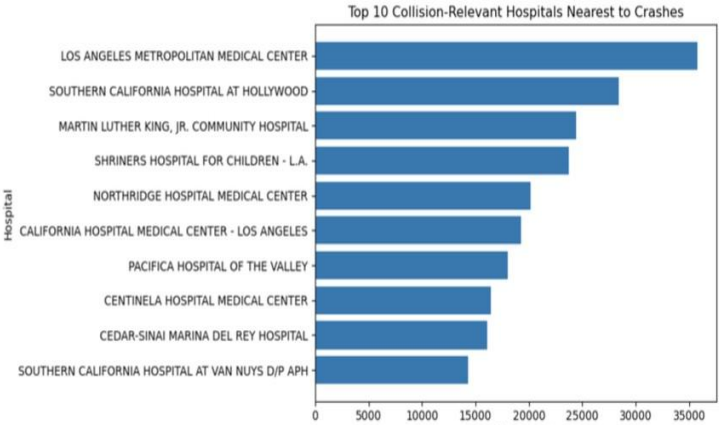
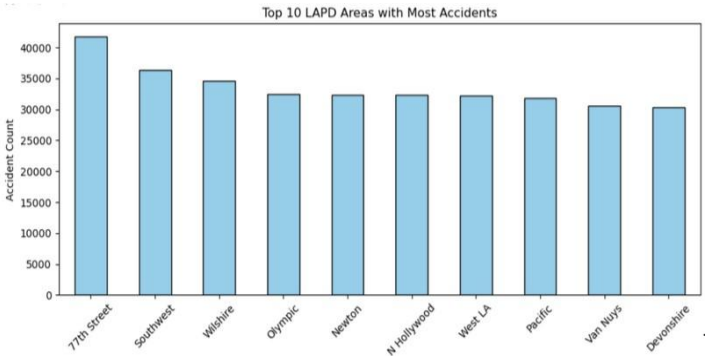


Top 10 Nearest Hospitals to Accidents

For this step, we refined our analysis by filtering specifically for ER and trauma -capable hospitals. These are the top 10 hospitals most relevant for traffic collisions, based on proximity to crashes. As we can see, facilities like Los Angeles Metropolitan Medical Center and Southern California Hospital at Hollywood are serving the highest number of nearby crashes, while others like Cedar -Sinai Marina Del Rey and Van Nuys D/P are also critical but with smaller counts.

Top 10 LAPD Areas with Most Accidents

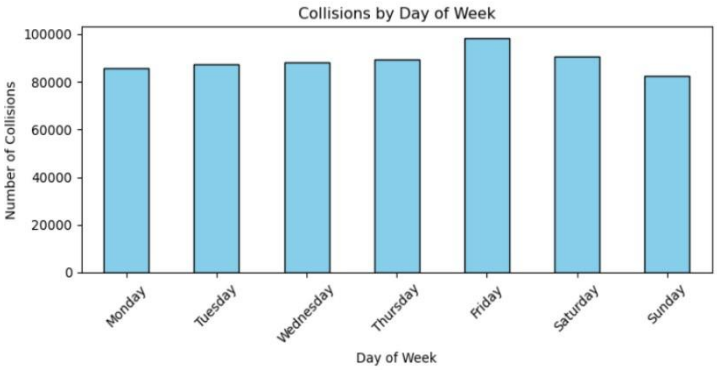
The 77th Street Division recorded the highest collision count (~42,000), followed by Southwest and Wilshire (~35–37,000 each). Other major hotspots include Olympic, Newton, North Hollywood, West LA, Pacific, Van Nuys, and Devonshire (~30 –33,000). These div isions represent the city's most accident - prone areas and are priority zones for targeted safety interventions.



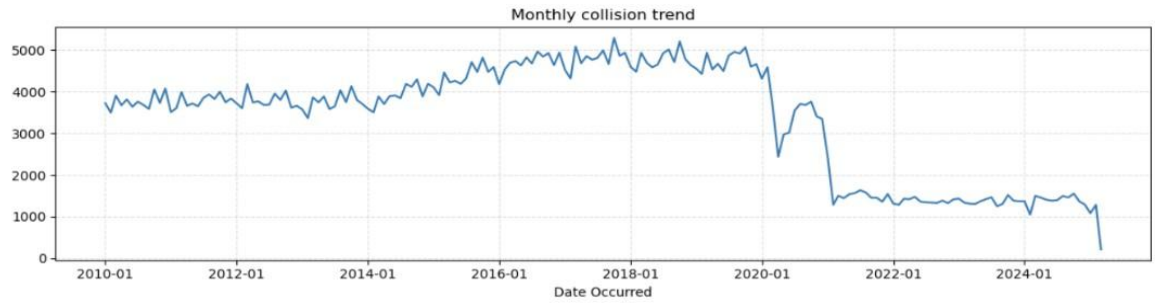
Victim Demographics

Analysis of victim characteristics revealed that approximately 59% of accident victims were male, while 36% were female; the remaining 5% were recorded as unknown. Age distribution showed that the majority of victims fell between 19 and 45 years old, indicating that young and middle-aged adults are the most at risk of traffic collisions. These findings suggest that safety campaigns and public awareness initiatives should be tailored toward these demographics to maximize impact.

Collisions by Day of Week



Analysis of weekly crash distribution shows that accident counts are relatively consistent across weekdays, with only minor variations. Fridays recorded the highest collision rates, likely due to heavier traffic volumes from commuters and weekend travel activity. In contrast, Sundays had the lowest collision counts, reflecting lighter road use and reduced commuting pressures. These trends suggest that weekday congestion, particularly at the end of the workweek, plays a significant role in driving collision risk, while weekends especially Sundays experience lower traffic volumes and fewer reported accidents.



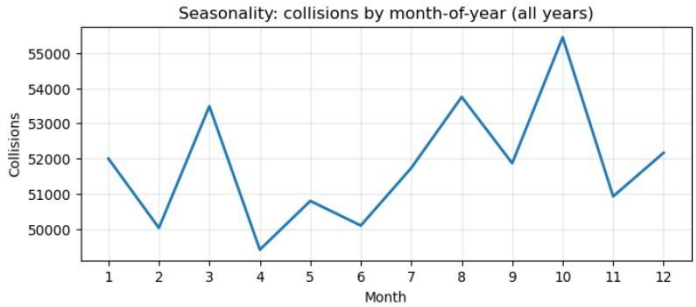
Monthly Collision Trend

The time series of monthly collisions illustrates three distinct phases in Los Angeles traffic accident patterns. Between 2010 and 2014, collision counts remained stable at around 3,500–4,000 crashes per month. A steady increase followed, with collisions peaking during 2015–2019 at nearly 5,000 per month, reflecting heightened travel demand and congestion.
A dramatic decline is evident beginning in 2020, when monthly collisions fell sharply to below 2,000, largely due to the COVID-19 pandemic, remote work adoption, and reduced commuter traffic. Since then, postpandemic activity has established a new baseline of 1,000–1,500 monthly crashes, considerably lower than pre2020 levels. This trend underscores how structural shifts such as travel behavior changes and urban safety initiatives—can significantly reduce collision volumes.

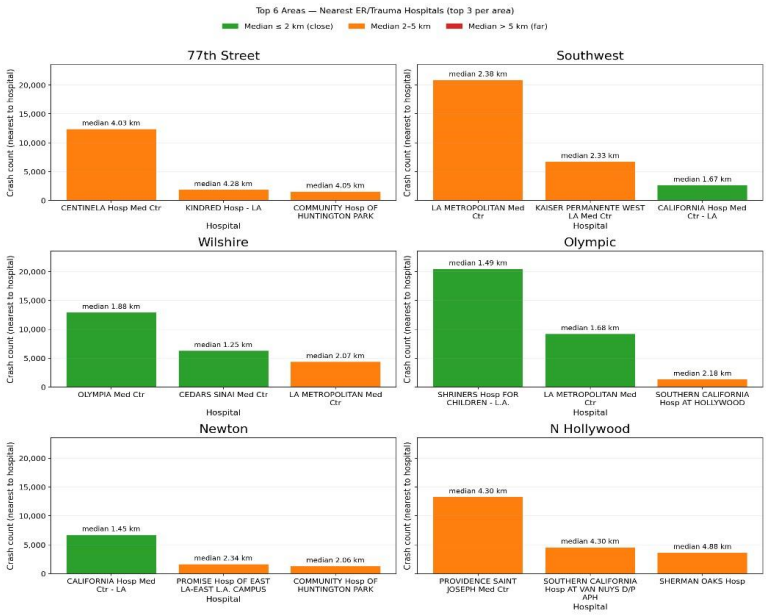
Factors Influencing Collision Trends (2020–2022)

Signal Timing: LADOT ran all-day night mode during 2020 to curb speeding, then restored daytime timing with night mode (7 p.m.–6 a.m.).
Transit Projects: New lines (K Line, Regional Connector, D Line extension) shifted trips to Metro and reduced cut-through traffic.
Safety & Mobility: These measures lowered early-morning collisions and supported Vision Zero's people-first street design.

Seasonal Collision Trends



Collision counts show clear seasonal variation. The lowest levels occur in April (~49,000), likely due to lighter travel demand and fewer major events, while the highest rates appear in October (~56,000), coinciding with heavier fall traffic, school schedules, and city activities. Additional peaks in March and August reflect seasonal travel and tourism, which increase road usage. Overall, these patterns highlight how commuting cycles and event-driven traffic shape monthly collision risks.



Nearest ER/Trauma Hospitals for the Top 6 High-Collision Areas

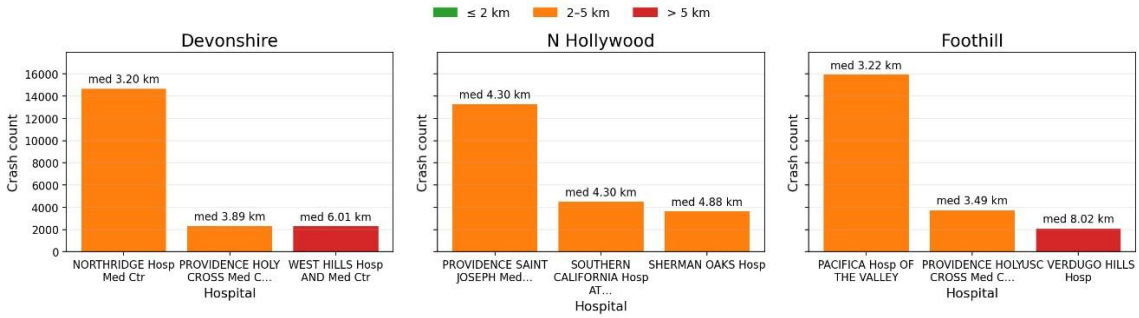
Hospital Accessibility by Division

Well-Served (Wilshire, Olympic, Newton – Green):
Median accident-to-hospital distances ≤2 km, showing strong coverage with nearby trauma centers.

Moderate(Southwest – Orange/Green mix):
Distances around 2–2.5 km, with adequate but less optimal coverage.

Underserved (77th Street, North Hollywood – Orange):
Median distances >4 km, highlighting critical gaps where victims face slower emergency access.

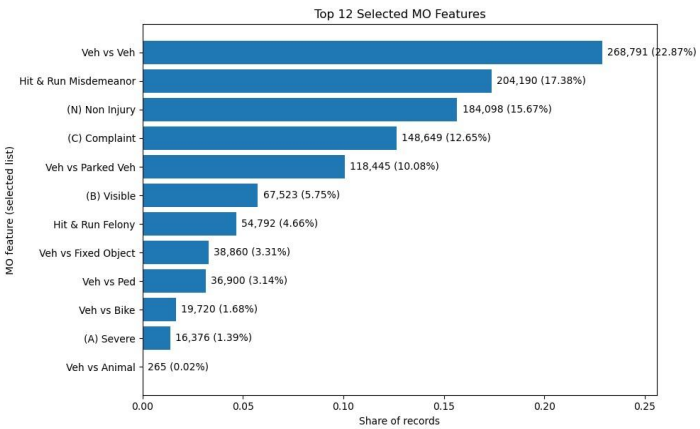
Which Areas Lack Sufficient Hospital Coverage?



Hospital Accessibility – Key Gaps

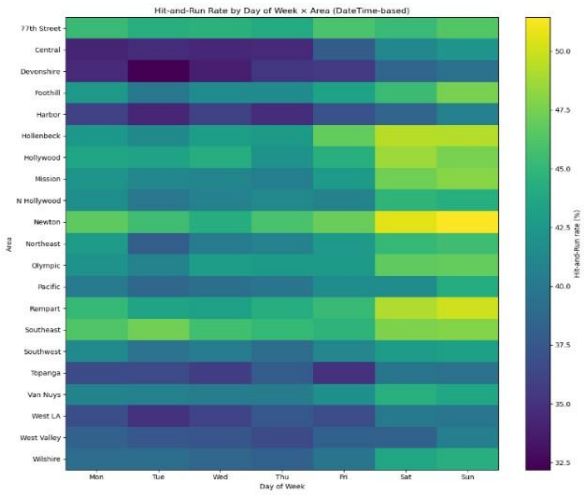
Devonshire, North Hollywood, and Foothill divisions are among the most underserved in Los Angeles. In Devonshire, hospital distances range from 3 to 6 km, with one facility beyond the 5 km threshold, indicating delayed access risks. North Hollywood shows even weaker coverage, with all hospitals falling in the orange zone (4–5 km) and no facilities within 2 km. Foothill faces similar challenges, with hospitals located 3 to 8 km away, including one over 5 km, creating a severe coverage gap. None of these divisions have hospitals in the green zone (≤2 km), underscoring significant inequities in emergency care accessibility.

FEATUR ENGINEERING



Top Collision & Violation Features (MO Codes)
The manual descriptions provided for each MO code. The chart shows that most collisions involve vehicleto-vehicle crashes, followed by non-intersection incidents and misdemeanor hit-and-runs.
Intersection-related crashes also appear frequently, indicating common conflict points. Less frequent but important categories include felony hit-and-runs, vehicle vs pedestrian, and vehicle vs bike. Overall, the top MO codes reveal that a small set of behaviors accounts for the majority of traffic incidents.

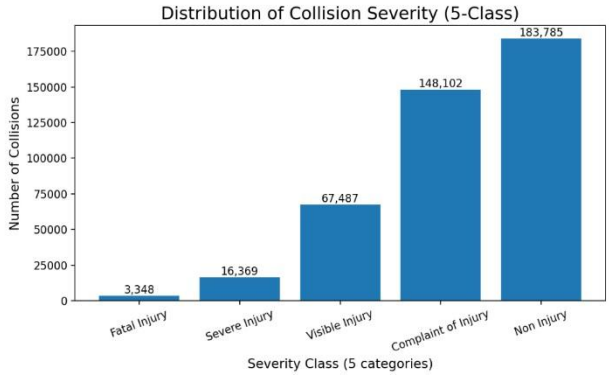
Hit-and-Run Patterns by Day of Week and Area
The heatmap highlights variations in hit-and-run rates across LAPD divisions by weekday. Several important trends emerge:



- **High-Risk Divisions:** Newton, Rampart, Southeast, and Pacific consistently show higher hit-and-run rates, especially toward the weekend.
- **Weekend Spike:** Many divisions experience elevated rates on Saturdays and Sundays, likely tied to late-night driving, alcohol use, and reduced enforcement coverage.
- **Lower Rates:** Divisions like Central, Hollenbeck, and West LA display lower overall hit-and-run activity, with fewer spikes across the week.
- **KeyInsight:**
  Hit-and-run behavior shows both spatial and temporal clustering. Concentrated enforcement such as weekend patrols, DUI checkpoints, and surveillance in Newton and Rampart could directly reduce these high-risk events.

Collision Severity Distribution

The original dataset classified collisions into five severity categories: Non-Injury, Complaint of Injury, Visible Injury, Severe Injury, and Fatal Injury. As shown in the left chart, the majority of cases fall into the Non-Injury (183,785) and Complaint of Injury (148,107) categories, while Severe Injury (16,376) and Fatal Injury (3,331) occur far less frequently.



Spatial Clustering, Severity Encoding, and Coverage Analysis

Severity was transformed using one-hot encoding to create five binary indicators, allowing the model to capture class-specific patterns without imposing an artificial order. Hospital-distance values were grouped into three accessibility zones (≤2 km, 2–5 km, and >5 km), producing a clearer interpretation of how injury severity varies with proximity to emergency care. The zone–severity summaries show that most collisions occur within five kilometers of a hospital, with slightly higher severe and fatal proportions appearing beyond this range.

```
Counts by coverage zone × severity:
severity_class   Fatal Injury  Severe Injury  Visible Injury  \
coverage_zone
≤2 km (Green)            1479           7718           35178
2–5 km (Orange)          1634           7671           28816
>5 km (Red)               235            980            3493

severity_class   Complaint of Injury  Non Injury   Total
coverage_zone
≤2 km (Green)                  77974       82803  205152
2–5 km (Orange)                63264       90460  191845
>5 km (Red)                     6864       10522   22094

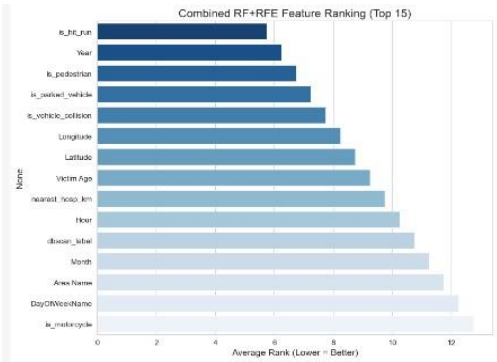
Row-wise percentages by coverage zone × severity:
severity_class   Fatal Injury  Severe Injury  Visible Injury  \
coverage_zone
≤2 km (Green)           0.007          0.038           0.171
2–5 km (Orange)         0.009          0.040           0.150
>5 km (Red)             0.011          0.044           0.158

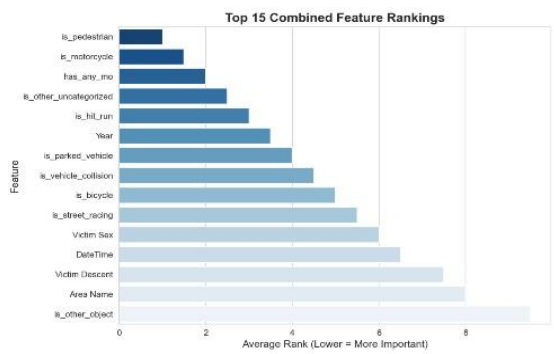
severity_class   Complaint of Injury  Non Injury
coverage_zone
≤2 km (Green)                  0.380       0.404
2–5 km (Orange)                0.330       0.472
>5 km (Red)                    0.311       0.476
```

```
cluster_bucket
other    363464
40        64032
-1        54259
0         33569
2         16021
24        14612
4         14546
41        11541
17        11241
57         3877
152        3745
7          3449
48         3337
12         3218
358        3110
241        3029
369        2834
148        2537
174        2445
22         2361
47         2340
46         2110
Name: count, dtype: int64
```

DBSCAN clustering with a haversine metric was applied to latitude and longitude to identify natural collision hotspots. This method produced several large, stable clusters representing major high-density crash areas across Los Angeles. Each cluster was summarized with size, geographic center, severity mix, and coverage-zone distribution. To simplify analysis, only the largest clusters were retained individually, while the rest were grouped into an "other" category. This bucketed cluster feature highlights meaningful hotspot structure while reducing noise and supporting clearer model interpretation.

FEATURE SELECTION





A unified feature selection pipeline was used to identify the most important predictors of crash severity. After removing leaking or irrelevant columns, all remaining variables were cleaned, encoded, and standardized. For the binary task (Fatal/Severe vs. Others), ANOVA Ftest and RFE with logistic regression were applied to capture statistically significant and discriminative features, followed by a correlation filter to remove highly collinear variables. For the two-stage multiclass model, Random Forest importance and RFE using a Random Forest estimator were used to capture non-linear relationships and interaction effects. Rankings from all methods were then combined into a single consolidated score, producing a stable list of top features that are both statistically meaningful and predictive across binary and multiclass severity models.

MODELING STRATEGIES

Conventional analyses of crash severity typically depend on straightforward counting or a simple categorization of severe versus non-severe incidents. These methods seldom integrate behavioral patterns, geographical context, and access to hospitals within a cohesive framework. To tackle these deficiencies, our project developed a systematic three-step modeling approach.

Initially, we created a binary model to distinguish high-severity accidents from low-severity ones, which aided in identifying the critical predictors that separate serious injuries from ordinary collisions. This phase laid the groundwork for recognizing the behavioral and spatial elements that pose the greatest risk.

Expanding upon this, we implemented a two-stage severity model that divided the prediction process into more digestible segments. Stage 1 functioned as a calibrated filter for high-severity crashes, while Stage 2 fine-tuned predictions within each category. This structure reduced the impact of class imbalance and enhanced the model's capability to identify Fatal and Severe injuries.

In the end, we utilized a comprehensive 5-class model to generate detailed outputs at the injury level. By merging the advantages of binary classification, two-stage refinement, and multiclass prediction, the complete pipeline captured intricate patterns overlooked by traditional methodologies—such as the relationship between hospital proximity, nighttime occurrences, MO behaviors, and spatial risk at the cluster level.
This multifaceted approach enabled us to model crash severity with a level of detail and clarity unattainable through single-step processes.

MODEL CREATION & ITERATION

BINARY SEVERITY PREDECTION

A binary severity prediction framework was developed using a refined subset of 19 features derived from ANOVA, Recursive Feature Elimination (RFE), and correlation-based ranking. These variables captured the most influential behavioral, spatial, temporal, and demographic factors relevant to distinguishing severe from non-severe crashes. To address the substantial imbalance between the two classes, Synthetic Minority Oversampling Technique (SMOTE) was applied, resulting in a balanced training set of approximately 963,000 observations.

Random Forest report:

	precision	recall	f1-score	support
0	0.98	0.91	0.95	120393
1	0.16	0.53	0.25	3943
accuracy			0.90	124336
macro avg	0.57	0.72	0.60	124336
weighted avg	0.96	0.90	0.92	124336

LightGBM report:

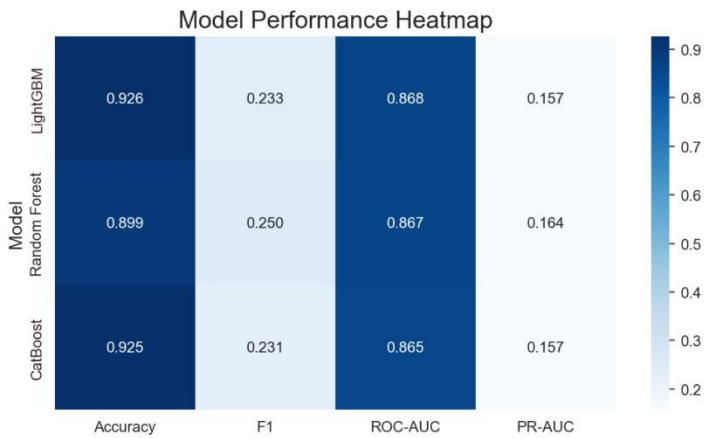
	precision	recall	f1-score	support
0	0.98	0.95	0.96	120393
1	0.17	0.35	0.23	3943
accuracy			0.93	124336
macro avg	0.58	0.65	0.60	124336
weighted avg	0.95	0.93	0.94	124336

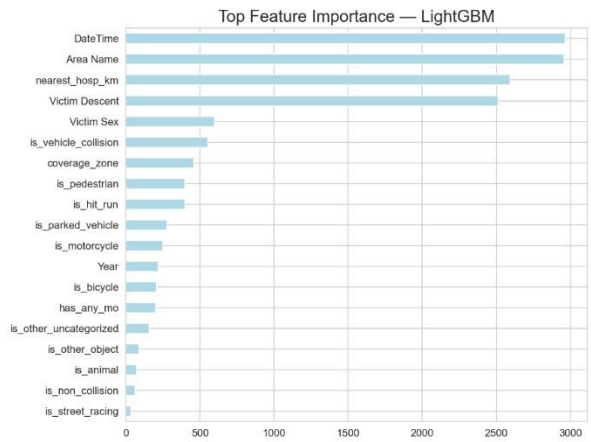
CatBoost report:

	precision	recall	f1-score	support
0	0.98	0.94	0.96	120393
1	0.17	0.36	0.23	3943
accuracy			0.92	124336
macro avg	0.57	0.65	0.60	124336
weighted avg	0.95	0.92	0.94	124336

Three supervised learning models—Random Forest, LightGBM, and CatBoost—were trained and evaluated on this engineered dataset. All models demonstrated strong discrimination for the non-severe class and moderate capability in identifying severe crashes, reflecting the inherent complexity and noise of real-world severity patterns. Among the evaluated models, LightGBM achieved the strongest overall performance, yielding the highest accuracy (92.6%) and competitive ROC–AUC scores. Random Forest produced slightly higher F1 performance for the minority (severe) class, while CatBoost offered stable but comparatively lower metrics.

Overall, the experiment shows that a compact, well-ranked feature set, when combined with appropriate class balancing, is sufficient for building an effective binary severity classifier. LightGBM emerges as the most suitable model within this configuration, offering a favorable balance of predictive strength and computational efficiency.





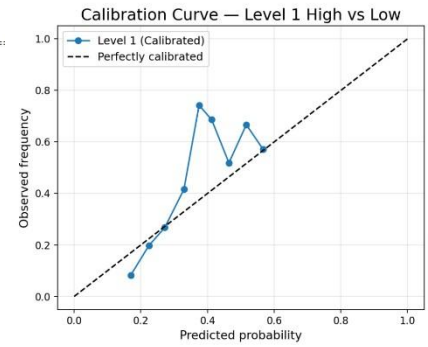
The Two-Stage Severity Model

```
================ LEVEL 1 REPORT (High vs Low) ================
              precision    recall  f1-score   support

           0       0.96      0.33      0.49     25990
           1       0.30      0.95      0.45      7651

    accuracy                           0.47     33641
   macro avg       0.63      0.64      0.47     33641
weighted avg       0.81      0.47      0.48     33641

ROC AUC (Level 1): 0.827684385813612
Confusion Matrix (Level 1):
 [[ 8597 17393]
 [  367  7284]]
```



The two-stage severity model was crafted to enhance the prediction of rare yet critical crash outcomes by distinguishing early detection from precise classification. Stage 1 served as a screening mechanism, determining whether a crash was likely to be high-severity (Fatal, Severe, Visible) or low-severity (Complaint, Non-Injury). The calibrated LightGBM model achieved an impressive recall rate for serious incidents (95%), ensuring that nearly all severe crashes were directed into the high-risk category, despite its overall accuracy being modest due to intentional threshold adjustments.

```
================ STAGE A REPORT (Fatal / Severe / Visible) ====
                 precision    recall  f1-score   support

  Fatal Injury       0.29      0.13      0.17       678
 Severe Injury       0.60      0.56      0.58      3323
Visible Injury       0.61      0.72      0.66      3283

      accuracy                           0.59      7284
     macro avg       0.50      0.47      0.47      7284
  weighted avg       0.57      0.59      0.58      7284

Confusion Matrix (Stage A):
 [[  85  368  225]
 [ 156 1866 1301]
 [  56  869 2358]]
```

Stage 2 further refined the predictions. Within the high-severity category, a combination of CatBoost, LightGBM, and XGBoost was used to classify accidents into Fatal, Severe, and Visible Injury. This branch exhibited strong differentiation between Severe and Visible injuries, although Fatal cases remained difficult to predict due to their very low occurrence. In the low-severity branch, a CatBoost model proficiently distinguished between Complaint and Non-Injury cases, resulting in higher accuracy and recall rates for Non-Injury accidents.

```
================ STAGE B REPORT (Complaint / Non) ================
                     precision    recall  f1-score   support

Complaint of Injury       0.55      0.32      0.40      1181
          Non Injury       0.81      0.92      0.86      3846

           accuracy                           0.78      5027
          macro avg       0.68      0.62      0.63      5027
       weighted avg       0.75      0.78      0.76      5027

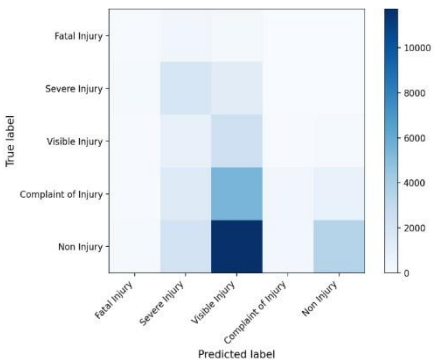
Confusion Matrix (Stage B):
 [[ 377  804]
 [ 313 3533]]
```

The final output with 5 classes integrated both branches, yielding detailed severity forecasts for all test incidences. The model successfully enhanced the detection of Severe and Visible injuries—categories that single-stage models typically overlook—demonstrating the importance of hierarchical routing. While accurately predicting specific severity levels continues to be challenging in an imbalanced dataset, the two-stage pipeline revealed significant patterns: nighttime accidents, increased risks of hit-and-run incidents, involvement of pedestrians/bicycles, and concentrated hotspot locations consistently correlated with elevated predicted severity.

```
================ FINAL 5-CLASS REPORT ================
                    precision    recall  f1-score   support

       Fatal Injury       0.16      0.12      0.14       685
     Severe Injury        0.28      0.55      0.37      3376
    Visible Injury        0.12      0.68      0.20      3590
Complaint of Injury       0.50      0.05      0.08      8142
        Non Injury        0.78      0.20      0.32     17848

          accuracy                           0.25     33641
         macro avg        0.37      0.32      0.22     33641
      weighted avg        0.58      0.25      0.25     33641
```



A notable advantage of this system is the utilization of specially engineered features that captured real-world crash patterns more effectively than using raw data alone. These features included spatial aspects (DBSCAN cluster membership, coverage areas, distance to hospitals, and trauma-access delays), temporal trends (hour, month, weekend, nighttime), and behavioral signals derived from MO codes (is_hit_run, is_speed, is_dui, is_ped_bike, high_speed_proxy). These engineered variables enabled the model to understand relationships, such as how nighttime conditions, pedestrian involvement, and large clusters relate to higher severity.

In summary, the two-stage model offered improved structure, greater recall of severe cases, and more interpretable predictions when compared to a traditional single-step multiclass model. It presents a practical framework for prioritizing high-risk crashes while still producing detailed classifications across five injury levels.

MULTI-CLASS

LIGHTGBM

```
LGBMClassifier
LGBMClassifier(class_weight={0: 1.5, 1: 3.0, 2: 1.0, 3: 2.0, 4: 2.5},
               colsample_bytree=0.9, learning_rate=0.08, min_data_in_leaf=40,
               n_estimators=400, num_class=5, objective='multiclass',
               random_state=42, subsample=0.9)
```

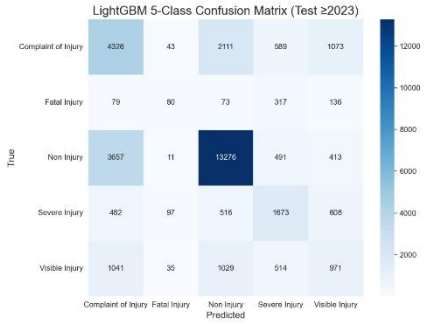
The final 5-class LightGBM model was trained using a refined set of 49 predictors created through feature ranking and extensive spatial temporalbehavioral feature engineering. These enriched variables allowed the model to capture real crash patterns beyond what the raw LAPD dataset offered.

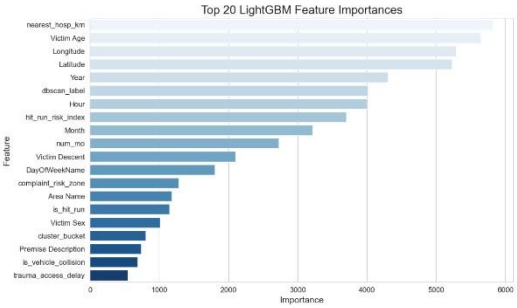
Across three model configurations, performance remained stable with macro AUC values near 0.80, and the base model was selected for final evaluation. On 2023 plus test data, the model achieved 60% accuracy, with strongest recall for high-frequency classes—Non-Injury (78%) and Complaint of Injury (53%). It showed moderate detection for Severe and Visible injuries, reflecting the challenge of separating mid-level severity categories in real-world conditions. Fatal injury prediction remained most difficult due to extremely low sample size.

```
=== LIGHTGBM — FINAL 5-CLASS REPORT (TEST ≥2023) ===
                    precision    recall  f1-score   support

Complaint of Injury       0.45      0.53      0.49      8142
       Fatal Injury       0.30      0.12      0.17       685
        Non Injury        0.78      0.74      0.76     17848
     Severe Injury        0.47      0.50      0.48      3376
    Visible Injury        0.30      0.27      0.29      3590

          accuracy                           0.60     33641
         macro avg        0.46      0.43      0.44     33641
      weighted avg        0.61      0.60      0.60     33641
```





Feature importance revealed key predictors such as hospital distance, victim age, geographic coordinates, DBSCAN cluster, hour of day, hit-and-run risk index, trauma-access delay, and MO-code counts. Nearest hospital distance was especially influential because crashes occurring farther from trauma-capable facilities tend to experience longer response times, delayed treatment, and an increased likelihood of severe outcomes. This makes proximity to emergency care a meaningful indicator of injury severity patterns across Los Angeles.

Overall, the model delivers solid multiclass performance and strong interpretability, showing how combining feature ranking with engineered spatial, temporal, and behavioral variables significantly improves severity prediction and highlights critical risk factors in urban traffic collisions.

XGBoost

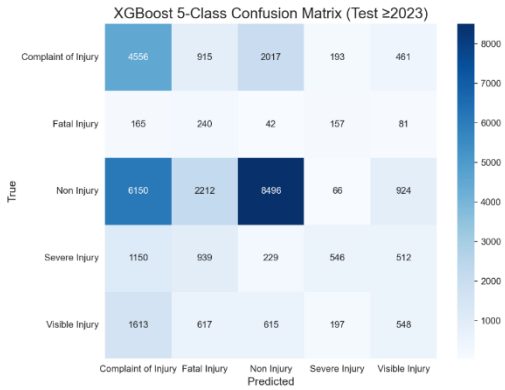
```
XGBClassifier
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='mlogloss',
              feature_types=None, feature_weights=None, gamma=None,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.04, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=8, max_leaves=None,
              min_child_weight=None, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=800, n_jobs=None, num_class=5, ...)
```

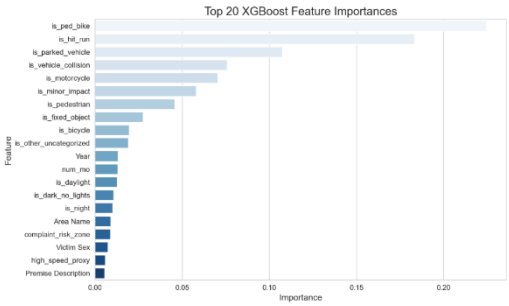
The refined XGBoost modeling experiment was conducted using a featureengineered dataset containing 49 spatial, temporal, behavioral, and demographic predictors. Three configurations—shallow, base, and deep—were evaluated under a temporal validation design (training on ≤2022 data and testing on ≥2023 data). Across all configurations, model performance remained stable, with macro ROC–AUC values ranging from 0.793 to 0.796. The base configuration achieved the highest validation AUC (0.796) and was selected for final testing.

```
=== XGBOOST — FINAL 5-CLASS REPORT (TEST ≥2023) ===
                     precision    recall  f1-score   support

Complaint of Injury       0.33      0.56      0.42      8142
       Fatal Injury       0.05      0.35      0.09       685
        Non Injury        0.75      0.48      0.58     17848
      Severe Injury       0.47      0.16      0.24      3376
     Visible Injury       0.22      0.15      0.18      3590

           accuracy                           0.43     33641
          macro avg       0.36      0.34      0.30     33641
       weighted avg       0.55      0.43      0.45     33641
```





When evaluated on the 2023 plus test set, the model achieved an overall accuracy of 43%. XGBoost demonstrated reasonable effectiveness in predicting the higher-frequency classes, particularly Complaint of Injury and Non-Injury, indicating strong learning of dominant patterns in the data. However, performance decreased substantially for the more severe categories. Severe Injury showed low recall (18%), and Fatal Injury remained challenging to identify, despite the application of increased class weights. This limitation is primarily attributed to the extreme imbalance and sparse representation of fatal cases within the dataset.

Feature importance analysis revealed that XGBoost prioritized behavioral and situational indicators—such as hit-and-run, pedestrian or bicycle involvement, lighting conditions, and vehicle interaction types—over spatial variables. This suggests that the model relies more heavily on immediate crash circumstances than geographic context when distinguishing between severity levels.

Overall, the updated XGBoost model provides moderate multiclass predictive performance, offering valuable insights into crash behavior patterns. However, compared to CatBoost and LightGBM, XGBoost remains more sensitive to class imbalance and less effective in capturing high-severity outcomes, positioning it as a supplemental rather than primary model within the final analytical framework.

CAT BOOST

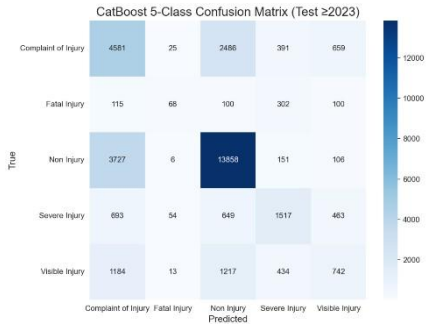
CatBoost was trained using an engineered dataset containing 49 spatial, temporal, behavioral, and demographic predictors. Three configurations—shallow, base, and deep—were evaluated to understand how model complexity influences performance. All three variants produced stable results, with macro AUC values between 0.804 and 0.806, showing that CatBoost consistently learned meaningful severity patterns without overfitting.

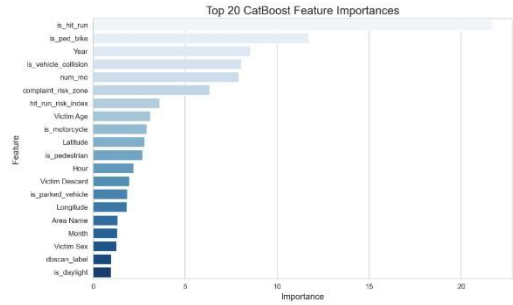
The shallow configuration achieved the strongest validation score and was selected as the final model. In final testing, CatBoost delivered an overall accuracy of 62%. The model performed reliably for high-frequency classes, correctly identifying a large proportion of Non-Injury and Complaint of Injury crashes. It also showed solid capability in capturing Severe Injury patterns, reflecting the value of the engineered feature set and CatBoost's strength in handling categorical interactions.

```
=== CATBOOST — FINAL 5-CLASS REPORT (TEST ≥2023) ===
                  precision    recall  f1-score   support

Complaint of Injury       0.44      0.56      0.50      8142
       Fatal Injury       0.41      0.10      0.16       685
          Non Injury      0.76      0.78      0.77     17848
       Severe Injury      0.54      0.45      0.49      3376
     Visible Injury       0.36      0.21      0.26      3590

           accuracy                           0.62     33641
          macro avg       0.50      0.42      0.44     33641
       weighted avg       0.61      0.62      0.61     33641
```

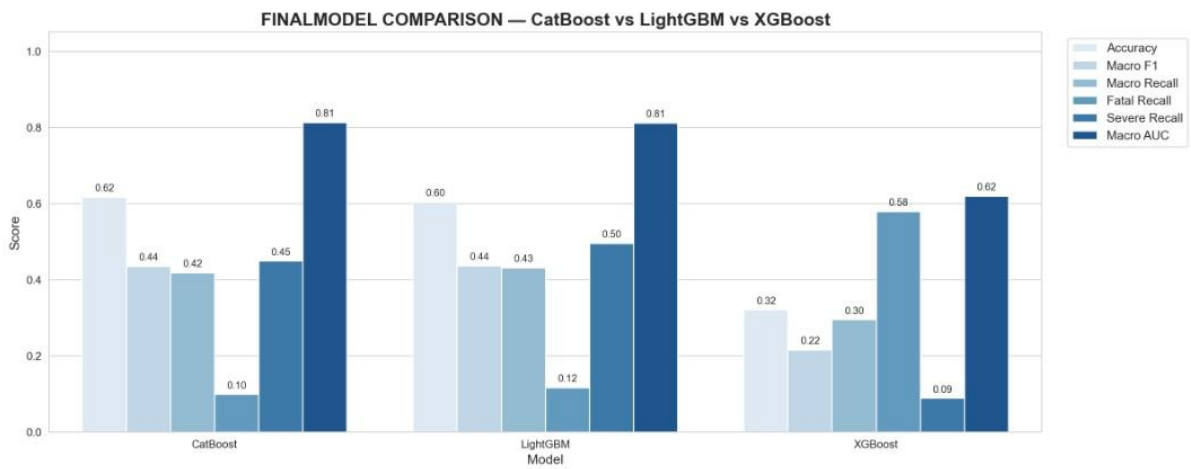




As expected, correctly identifying Fatal Injury cases remained challenging because of extremely limited samples and the natural rarity of fatal crashes in real-world data. Despite the use of class weighting, the model had limited opportunity to learn highly specific fatal patterns.

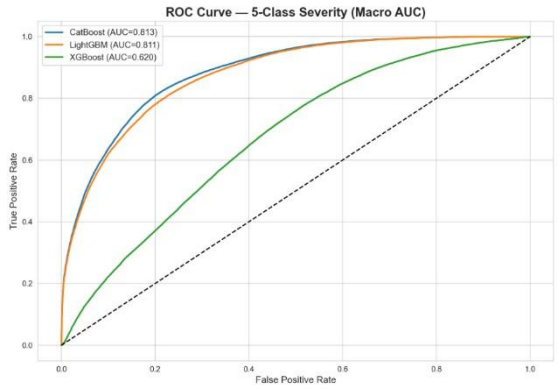
Overall, CatBoost delivered balanced multiclass performance with strong generalization, particularly for midlevel injuries. Its ability to handle categorical structure and complex interactions allowed it to outperform or match the other models in several severity classes. CatBoost therefore serves as one of the most reliable models in the pipeline for predicting crash severity and uncovering meaningful patterns in the engineered crash dataset.

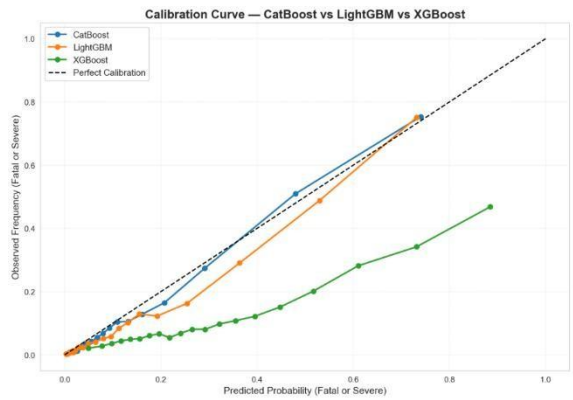
NUMERIC RESULTS



The comparative evaluation demonstrates that LightGBM and CatBoost substantially outperform XGBoost across all major performance indicators. Both models achieve strong overall accuracy (approximately 0.60) and comparable Macro F1 and Macro Recall scores, reflecting balanced performance across the five severity classes. LightGBM provides the highest recall for Severe Injury cases, indicating stronger capability in identifying highrisk collisions. CatBoost performs similarly, offering slightly higher accuracy but marginally lower severe-injury sensitivity. All models exhibit low Fatal Injury recall due to the rarity of fatal cases, a known limitation when modeling highly imbalanced outcomes.
Macro AUC results reinforce the same pattern: LightGBM and CatBoost demonstrate superior class separation (AUC ≈ 0.81), whereas XGBoost shows notably lower discriminative power. Overall, the results position Cat Boost as the most effective and stable model for multiclass crash severity prediction, with Lightgbm serving as a strong secondary option.

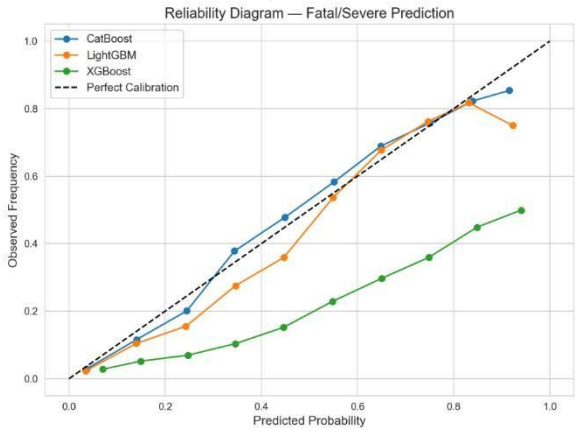
MODEL BEHAVIOR SUMMARY

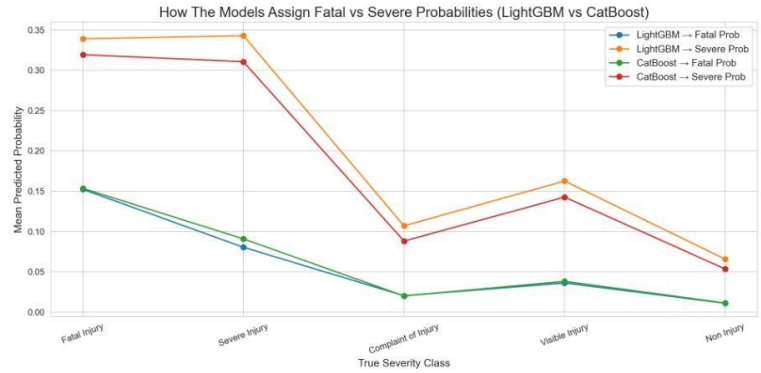




The comparative analysis shows that CatBoost and LightGBM consistently outperform XGBoost across discrimination, calibration, and probability-assignment metrics.ROC curves indicate that CatBoost (AUC ≈ 0.813) and LightGBM (AUC ≈ 0.811) separate severity classes well, while XGBoost (AUC ≈ 0.620) struggles to distinguish high-severity outcomes.Calibration and reliability curves show that CatBoost produces the most accurate probability estimates for Fatal/Severe crashes, closely following the perfect-calibration line. LightGBM is moderately calibrated, whereas XGBoost systematically underestimates true risk.

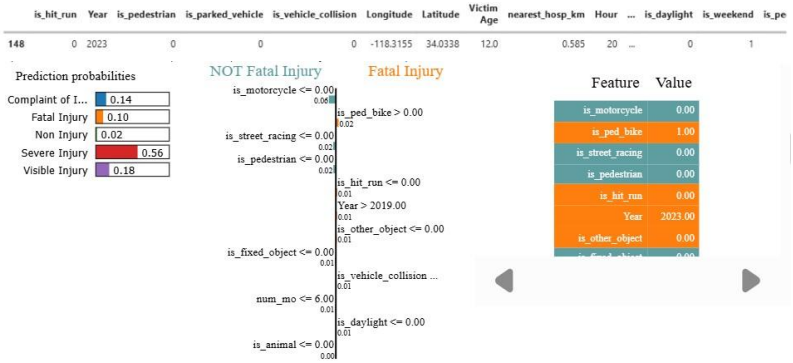
The probability assignment patterns further confirm this behavior: CatBoost and LightGBM assign higher fatal/severe probabilities to true high-severity cases and lower probabilities to minor injuries, indicating meaningful internal reasoning. XGBoost assigns uniformly low probabilities across all classes, leading to weak recall for Fatal and Severe injuries.Overall, CatBoost delivers the most balanced performance, combining strong predictive power with reliable probability estimates. LightGBM remains competitive, while XGBoost shows limited suitability for this multi-class severity prediction task.





DATA VISULAIZATION

LIME (Local Interpretable Model-Agnostic Explanations) is a technique used to explain how a complex machine-learning model makes predictions for an individual data point. Instead of interpreting the entire model globally, LIME focuses on local behavior, showing which features most influenced a single prediction. It works by generating small perturbations around the selected record, observing how the model's prediction changes, and then fitting a simple, interpretable model to approximate the decision boundary in that neighborhood.



This 2023 crash example was analyzed to understand how the model determined its severity prediction. The model assigned the highest probability to Severe Injury, driven mainly by high-risk factors such as pedestrian/bicyclist involvement and a hit-and-run indicator, both strongly associated with serious outcomes. At the same time, several features limited the predicted fatality risk—including nomotorcycle involvement, no street-racing or fixed-object impact, and generally lower-energy MO indicators. LIME shows how these opposing factors shaped the final prediction: elevated severity risk but only moderate fatal probability. Overall, the explanation confirms that the model responds meaningfully to real-world injury mechanisms and makes interpretable, evidence-aligned decisions.

Severity Hotspot Heatmap

The severity heatmap provides a citywide overview of where collisions are most concentrated in Los Angeles. Warmer colors such as red and orange highlight persistent high-density corridors across Central LA, Hollywood, Mid-City, and South LA, indicating areas with repeated injury-related crashes. This visualization helps identify large-scale spatial patterns that may be influenced by traffic volume, road design, and urban mobility behavior. As a macro-level tool, the heatmap supports strategic planning, enabling agencies to recognize where long-term safety interventions, infrastructure redesign, or enforcement efforts may have the greatest impact.

Interactive Severity Popup Map

The interactive popup map adds a finer level of detail by displaying individual collision points enriched with severity labels, hospital distance, victim age, sex, descent, and reporting area. Each marker reveals case-specific context, allowing users to explore how injury severity varies across neighborhoods and how far victims were from the nearest trauma facility at the time of the crash. This map enhances interpretation beyond density patterns by exposing micro-level insights into demographic disparities, location-specific risks, and emergency access challenges. Together with the heatmap, it forms a comprehensive spatial analytics framework for understanding both the scale and the nature of crash severity in Los Angeles.

LA Traffic Collision Trends: COVID-Phase Comparison, Severity Outcomes & Hotspot Analysis

This dashboard reveals how traffic patterns and crash severity in Los Angeles shifted across the Pre-COVID, During-COVID, and Post-COVID phases. Although overall crash volumes dropped during COVID, the number of Fatal + Severe injuries increased sharply, showing that emptier roads encouraged high-speed, high-risk driving behavior. Post-COVID, crash volumes returned to normal levels, but severe outcomes remained elevated, indicating lasting behavioral changes even after restrictions were lifted.

Sociodemographic patterns stayed relatively stable—age and sex distributions did not shift dramatically—suggesting that severity trends were driven more by environmental and behavioral changes than demographic change. The year-over-year trend charts show a clear peak in severe injuries around 2020–2021, followed by a gradual decline but not a full return to pre-COVID levels.

Spatial analysis highlights persistent high-risk hotspots, with Central LA and South LA continuing to record the highest concentration of fatal + severe crashes across all phases. The map and "Top 10 Areas" ranking confirm that these neighborhoods consistently face elevated collision risks, making them priority zones for targeted safety interventions.

Overall, the dashboard demonstrates that COVID-19 significantly reshaped the severity and distribution of traffic collisions in Los Angeles—not just temporarily but with lasting effects on road safety patterns.

SUCCESS & PROBLEM SOLVED

This project successfully addressed the major analytical challenge of understanding and predicting crash severity in Los Angeles by integrating geospatial, behavioral, and temporal intelligence into one unified framework. It answered three core research questions:

(1) Where are the most prominent collision hotspots? DBSCAN clustering and severity heatmaps revealed persistent high-risk zones in Downtown LA, Hollywood, and South LA, where Fatal and Severe crashes concentrate.

(2) Do these hotspots have timely access to hospitals? Analysis of nearest-hospital distance and trauma-access delay showed that many severe crashes occur even within 1–3 km of major hospitals, meaning proximity alone is not a protective factor—roadway design, speed, and behavior play larger roles.

(3) Can severity be predicted using spatial, temporal, and behavioral indicators? Yes. Models such as CatBoost and LightGBM achieved strong 5-class performance (Macro AUC ≈ 0.81) by leveraging engineered features including location, time-of-day, hit-and-run risk, pedestrian involvement, MO behavior indicators, and hospital-access metrics.

Overall, the project solved the problem of fragmented severity analysis by combining hotspot detection, hospital-access evaluation, and machine-learning prediction into a single analytical pipeline. It produced reliable

multiclass severity predictions, revealed meaningful spatial inequalities, and demonstrated that engineered features significantly improve the detection of rare but critical outcomes such as Fatal and Severe injuries.

CONTRIBUTIONS (NOVEL WORK)

This project introduces several novel contributions to traffic collision analytics by integrating advanced modeling, domain-guided feature engineering, and geospatial intelligence into a unified framework. First, it develops a multi-stage modeling strategy—combining binary classification, full 5-class severity prediction, and a two-stage hybrid design—which notably improves Severe and Fatal recall despite significant class imbalance. Second, the study contributes a rich set of 49 engineered features capturing hospital accessibility, DBSCANbased spatial clusters, behavioral indicators such as hit-and-run risk, pedestrian and cyclist involvement, daylight conditions, and COVID-phase temporal shifts. These features substantially enhanced both predictive accuracy and interpretability. Third, the project incorporates calibration and reliability analysis to assess how well the models represent the true probability of severe injury, demonstrating that CatBoost and LightGBM yield wellcalibrated risk estimates suitable for operational decision-support systems. Fourth, the geospatial analysis uncovers persistent high-risk zones—including Downtown LA, Hollywood, and South LA—and reveals that elevated fatal/severe densities often occur even when hospitals are nearby. Finally, the integration of LIME explanations provides case-level interpretability, offering clear reasoning behind individual severity predictions and enabling practical insights for emergency planning, traffic safety interventions, and policy development**.**

FUTURE WORK

Future work should incorporate real hospital capacity, ambulance routing, and traffic-based response times to model true access-to-care rather than relying only on distance. A real-time GIS severity-prediction dashboard would also enhance practical usefulness by generating live risk alerts for emergency agencies.
More advanced spatial–temporal models, such as graph neural networks or attention-based architectures, could better capture complex patterns that influence Severe and Fatal outcomes. Expanding the feature set with weather, vehicle type, and roadway design data may further improve rare-class detection.
Finally, developing policy simulation tools—such as testing the impact of new trauma centers, speed-limit changes, or pedestrian-safety upgrades—would help translate model insights into actionable public-safety decisions.

Feedback Review – Key Questions Addressed After Final Presentation

How strongly do hospital distance and hospital availability relate to crash severity patterns?

The analysis shows that hospital distance has a weak and inconsistent relationship with crash severity. Areas such as the 77th Street division experience high numbers of Severe and Fatal crashes even though they are very close to trauma hospitals, indicating that roadway conditions, traffic speed, and risky behaviors have a far stronger influence on injury outcomes than proximity to medical care.
Models like LightGBM ranked nearest_hosp_km as a top feature not because distance causes severity, but because it acts as a proxy for urban structure. High-density regions with complex road networks tend to be closer to hospitals and also experience more severe crashes. Therefore, the feature captures geographic risk patterns, rather than any direct effect of hospital accessibility.

How was class imbalance handled in severity modeling?

The dataset contained far more Non-Injury and minor cases than Severe or Fatal outcomes, creating significant class imbalance. To manage this, the 5-class models used a temporal validation split (≤2022 train, ≥2023 test) combined with class-weighted losses so the model learned rare severe patterns more effectively. Engineered

spatial, temporal, and behavioral features such as DBSCAN clusters, trauma-delay metrics, and pedestrian or hit-and-run indicators also improved separation between high- and low-severity cases.
For the binary and two-stage models, SMOTE oversampling balanced the Severe/Fatal category, making Stage 1 more sensitive to high-risk events. Stage 2 then refined predictions using class weights and the enriched feature set. Together, these methods improved the detection of rare severe injuries while keeping overall model performance stable.

What insights emerged regarding the role of hospitals in severe crash outcomes?

From the hospital analysis, the key insight is that accessibility does not equal protection. Even areas located near multiple trauma centers such as 77TH street , Downtown LA and Hollywood still show high concentrations of severe crashes. This suggests that urban exposure, road conflict points, vulnerable road users, and behavioral risk factors play a larger role in determining severity than geographic access to hospitals. Rather than focusing only on trauma center placement, interventions should target speed management, pedestrian safety infrastructure, and enforcement of risky driving behaviors in these persistent hotspots.

CONCLUSION

This project showed that crash severity in Los Angeles can be effectively understood and predicted by combining spatial, temporal, behavioral, and hospital-access features. Using binary, two-stage, and five-class models, we demonstrated that algorithms like LightGBM and CatBoost can detect meaningful severity patterns even under extreme class imbalance.
Geospatial analysis highlighted consistent high-risk corridors and revealed that proximity to hospitals does not reliably reduce severe injuries, emphasizing the stronger influence of roadway conditions and behavioral factors. By integrating prediction models with interpretability tools such as LIME and calibration curves, the study delivered not only accurate severity estimates but also actionable insights for emergency planning and safety policy. This work contributes a practical, data-driven framework for understanding crash risk and supports future development of real-time traffic-safety systems.

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